Some patients had missing values for one or more predictor variables; in this case we imputed the missing data by using the median value in the sample, or for laboratory values, using the recommended default value listed on the Vanderbilt Biostatistics web site. If more than 4,000 patients were missing a value for the variable, that variable was excluded from analysis. After processing, there were 39 predictor variables. Patients were divided with a 70%/30% split into training and test sets. The processed dataset is available at our project's GitHub page.

We tested four models on the SUPPORT study dataset:

- Our model, Nnet-survival (flexible version, so that non-proportional hazards were possible)
- Cox-nnet (Ching et al., 2018)
- Deepsurv (Katzman et al., 2018)
- Standard Cox proportional hazards model

All three neural network models used a simple multilayer perceptron architecture with a single hidden layer. The Cox-nnet default parameters specify a hidden layer size of 7 neurons when input dimension is 39, which we felt to be a reasonable choice, so a hidden layer size of 7 was used for the three models. For all three neural network models, L2 regularization was used to help prevent overfitting. The regularization strength parameter was chosen using 10-fold cross validation on the training set, using log likelihood as the performance metric. No regularization was used for the standard Cox proportional hazards model. For Nnet-survival, 19 follow-up time intervals were used, extending out to 6 years (around the maximum follow-up time of the SUPPORT study), with larger spacing for later intervals due to the decreased density of failure events with increasing follow-up time. The RMSprop optimizer was used for Nnet-survival.

As Cox-nnet and Deepsurv only output a prognostic index for each patient, not a predicted survival curve, we generated predicted survival curves for these methods by using the Breslow method to generate a baseline hazard function (Breslow, 1974).

To evaluate the models' discrimination performance, we used Harrell's C-index to assess discrimination. To calculate C-index for Nnet-survival, the model's predicted probability of 1-year survival was used to rank the patients.

To evaluate model calibration, we used a published adaptation of the Brier score for censored data (Graf et al., 1999). This was implemented using the *ipred* R package. We also created calibration plots for specific follow-up times (Royston and Altman, 2013).

We tested the running time of each method by fitting each model using a range of dataset sizes. Simulated datasets ranging from 1,000 to 1,000,000 patients were created by sampling from the 9,105 SUPPORT study patients with replacement. Each combination of model and sample size was run three times and the results were averaged. Each model was trained for 1,000 epochs. An Ubuntu Linux server with 3.6GHz Intel Xeon E5-1650 CPUs and 32GB of RAM was used. The models were constrained to run on one CPU core. Python version 3.5.2 was used for the Nnet-survival, Cox-nnet, and standard Cox proportional hazards models; Python version 2.7.12 was used for Deepsurv. R version 3.4.3 was used to calculate Brier scores. The code for the SUPPORT study analysis is available in `support_study.py` in the GitHub repository.

## RESULTS

### Simulated data

#### *Simple model with one predictor variable*

We tested a simple survival model with one binary predictor variable and no hidden layers. The calibration curves for the two groups of patients are shown in Fig. 3. It can be seen that calibration is excellent: the actual and predicted survival curves are superimposed.

Model convergence was found to be reliable. The model was optimized repeatedly with different random starting weights and converged to very similar final loss/likelihood values.

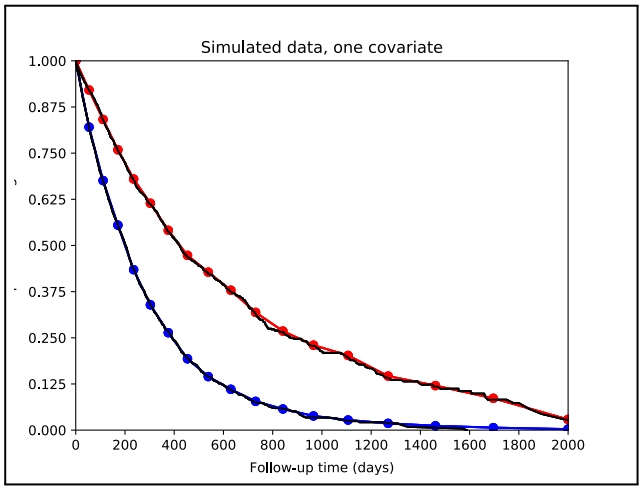



**Figure 3.** Simple survival model with one predictor variable. 5000 simulated patients with exponential survival distribution. Half of patients have predictor variable value of 0 with median survival of 200 days; the other half have value of 1 with median survival of 400 days. Actual survival for the two groups is shown in black (Kaplan-Meier curves). The average model predictions for the two groups are shown in blue and red, respectively. Model predictions correspond well to actual survival.

### *Optimal width of time intervals*

The discrimination performance of the survival model was robust to various choices of time interval width and configuration (constant width, or increasing width with increasing follow-up time). For each of the four time interval options, discrimination performance was identical with C-index of 0.66.

### *Convolutional neural network for MNIST dataset*

We used the MNIST dataset of handwritten digits to simulate a scenario where each patient has an X-ray image of a tumor, and survival time distribution depends on the appearance of the tumor. Digits 0 through 4 were used, with lower digits having longer median survival. The model's task was to accurately predict survival time for each patient. There were 30,596 images in the training set used to train the model's weights, and 5,139 in the test set used to evaluate model performance.

The Nnet-survival model produced good performance. C-index for the test set was 0.713, compared to 0.770 for a "perfect" model that used the true digit as the predictor variable. Calibration was excellent, as seen in Fig. 4.

## SUPPORT study (real data)

Four survival models (Nnet-survival, Cox-nnet, Deepsurv, and a standard Cox proportional hazards model) were tested using the SUPPORT study dataset of 9,105 hospitalized patients.

We found that several predictor variables violated the proportional hazards assumption of the standard Cox model, with an example given in Fig. 5. This provides an opportunity for our Nnet-survival model to have improved calibration compared to the other three models.

All models were trained/fit using the 70% of patients in the training set (n=6,373). Then, performance was measured using the remaining 30% of patients in the test set (n=2,732).

Discrimination performance was very similar for all models, with test set C-index around 0.73 (Table 1). Table 1 also shows calibration performance as measured by the Brier score (lower is better). Nnet-survival had the best calibration performance at all three follow-up time points, though the differences were fairly small. Calibration was also assessed visually using calibration plots (Fig. 6). Our Nnet-survival model appeared to have the best calibration at the 6 month and 1 year time points, with Cox-nnet and the standard Cox model tending to under-predict survival probability for the best-prognosis patients.

**Table 1.** Performance of four models on SUPPORT study test set (n=2,732). Discrimination assessed with C-index. Calibration assessed with modified Brier score (Graf et al. (1999)) at three specific follow-up times.

| Model | C-index | Brier score: 6 months | Brier score: 1 year | Brier score: 3 years |
|---|---|---|---|---|
| Nnet-survival | 0.732 | 0.181 | 0.184 | 0.177 |
| Cox-nnet | 0.735 | 0.183 | 0.185 | 0.177 |
| Deepsurv | 0.730 | 0.184 | 0.187 | 0.179 |
| Cox PH | 0.734 | 0.183 | 0.186 | 0.178 |

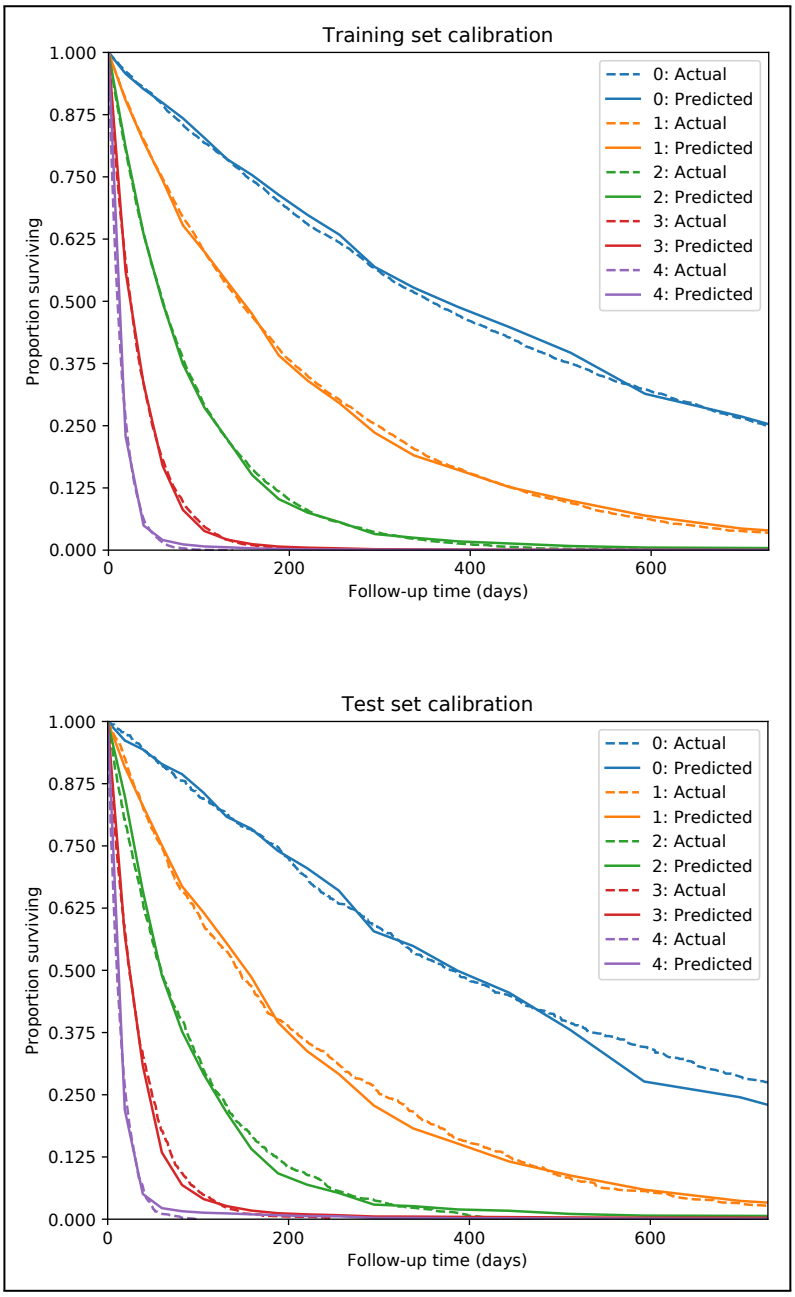



**Figure 4.** MNIST dataset calibration plots. Images of handwritten digits 0-4 were used as predictor of survival time. Lower digits have longer median survival. For each digit, actual survival curve plotted with dotted line and mean model-predicted survival plotted with solid line.